

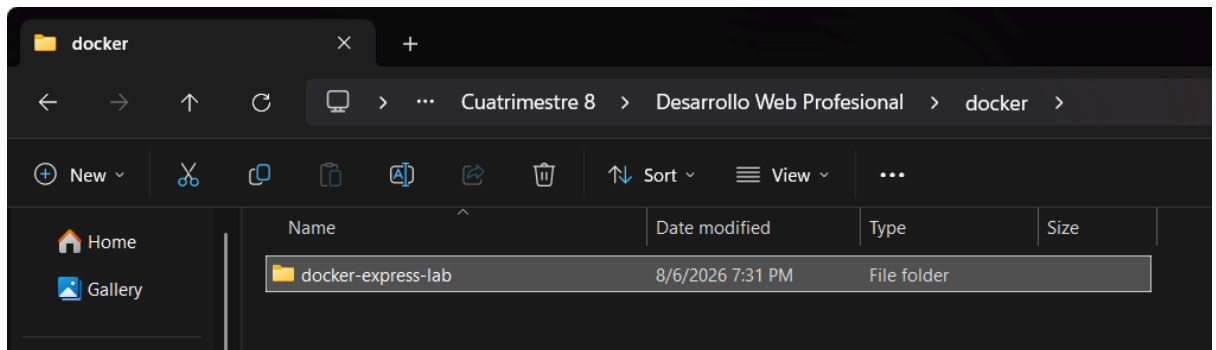
Ejercicio Docker

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IDGS 8-3

Parte 1: Estructura y Código del Servidor Web

1.



2.

```
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> npm init -y
Wrote to C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab\package.json:

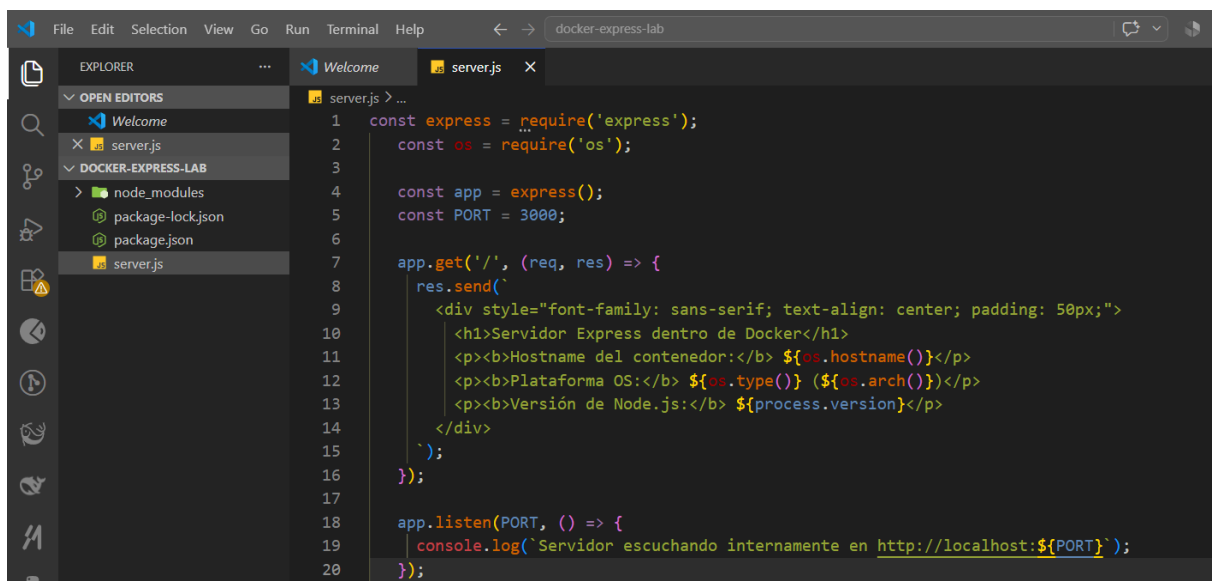
{
  "name": "docker-express-lab",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> npm install express
added 67 packages, and audited 68 packages in 5s

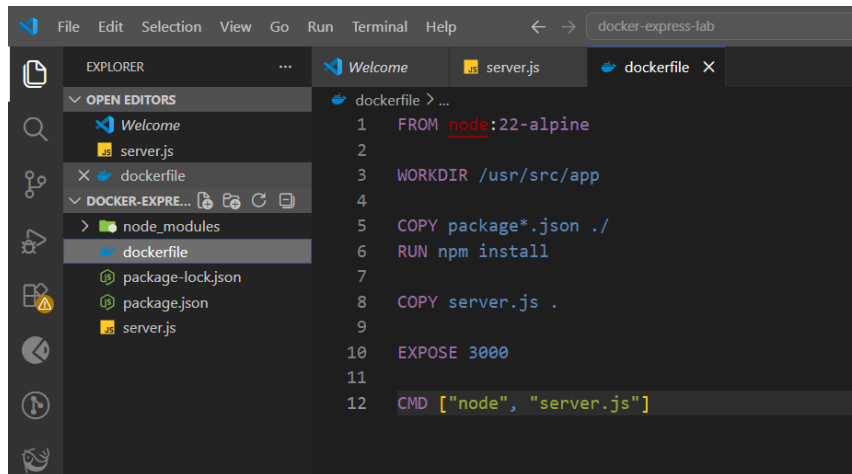
26 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> |
```

3.



Parte 2: Dockerfile con Gestión de Dependencias



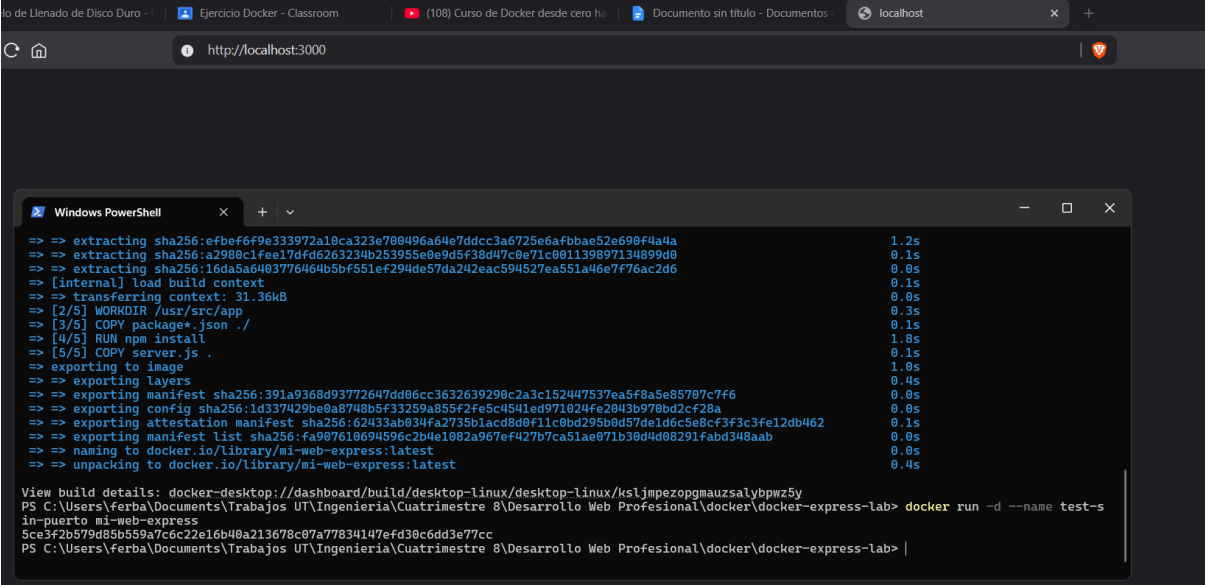
Parte 3: Construcción de la Imagen

```
found 0 vulnerabilities
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker build -t mi-web-express .
[+] Building 8.9s (4/9)
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 188B
=> [internal] load metadata for docker.io/library/node:22-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:22-alpine@sha256:c610fcd4b1d5b4740dd70c284ed3cb16bb857e0f7166196e36a5501df7a3aa32
=> => resolve docker.io/library/node:22-alpine@sha256:c610fcd4b1d5b4740dd70c284ed3cb16bb857e0f7166196e36a5501df7a3aa32 [+] Building 14.5s (10/10) FINISH
ED
=> [internal] load build definition from dockerfile
=> => transferring dockerfile: 188B
=> [internal] load metadata for docker.io/library/node:22-alpine
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/node:22-alpine@sha256:c610fcd4b1d5b4740dd70c284ed3cb16bb857e0f7166196e36a5501df7a3aa32
=> => resolve docker.io/library/node:22-alpine@sha256:c610fcd4b1d5b4740dd70c284ed3cb16bb857e0f7166196e36a5501df7a3aa32
=> sha256:16da5a609377646b5bf551ef294de57da242ae594527ea551a46e7f76ac2d6 405B / 405B
=> sha256:a2980c1fee17df6263234b253955e0e9d5f38d47c0e71c001139897134899d0 1.26MB / 1.26MB
=> sha256:efbef6f9e333972a10ca323e700496a64e7ddcc3a6725e6afbbae52e690f4a4a 52.63MB / 52.63MB
=> sha256:55afalecc21d2bb5e5045f32dafee562727fd89860bac26f6c32123439af26a4 3.85MB / 3.85MB
=> => extracting sha256:55afalecc21d2bb5e5045f32dafee562727fd89860bac26f6c32123439af26a4
=> => extracting sha256:efbef6f9e333972a10ca323e700496a64e7ddcc3a6725e6afbbae52e690f4a4a
=> => extracting sha256:a2980c1fee17df6263234b253955e0e9d5f38d47c0e71c001139897134899d0
=> => extracting sha256:16da5a609377646b5bf551ef294de57da242ae594527ea551a46e7f76ac2d6
=> [internal] load build context
=> => transferring context: 31.36kB
=> [2/5] WORKDIR /usr/src/app
=> [3/5] COPY package*.json ./
=> [4/5] RUN npm install
=> [5/5] COPY server.js .
=> => exporting to image
=> => exporting layers
=> => exporting manifest sha256:391a9368d93772647dd06cc3632639290c2a3c152447537ea5f8a5e85707c7f6
=> => exporting config sha256:1d337429be0a8748b5f33259a855f2fe5c4541ed971024fe2043b970bd2c2f8a
=> => exporting attestation manifest sha256:62433ab034fa2735b1acd8d0f11c0bd295b0d57deld6c5e8cf3f3c3fe12db462
=> => exporting manifest list sha256:fa907610694596c2b4e1082a967ef427b7ca51ae071b30d4d08291fabd348a8ab
=> => naming to docker.io/library/mi-web-express:latest
=> => unpacking to docker.io/library/mi-web-express:latest
```

4.

Parte 4: El Mapeo de Puertos (-p)

5.



```
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker build -t mi-web-express .
=> => extracting sha256:efbef6f9e333972a10ca323e700496a64e7ddcc3a6725e6afbbae52e690f4a4a 1.2s
=> => extracting sha256:a2980c1fee17d4fd6263234b253955e0e9d5f38d47c0e71c001139897134899d0 0.1s
=> => extracting sha256:16da5a6403776464b5bf551ef294de57da242eac594527ea551a46e7f76ac2d6 0.0s
=> [internal] load build context 0.1s
=> => transferring context: 31.36kB 0.0s
=> [2/5] WORKDIR /usr/src/app 0.3s
=> [3/5] COPY package*.json ./ 0.1s
=> [4/5] RUN npm install 1.0s
=> [5/5] COPY server.js . 0.1s
=> exporting to image 1.0s
=> => exporting layers 0.4s
=> => exporting manifest sha256:391a9368d93772647dd06cc3632639290c2a3c152447537ea5f8a5e85707c7f6 0.0s
=> => exporting config sha256:1d337429be0a8748b5f33259a855f2fe5c4541ed971024fe2043b970bd2cf28a 0.0s
=> => exporting attestation manifest sha256:62433ab034fa2735blacd8d0f11c0bd295b0d57del6dc5e8cf3f3c3fe12db462 0.1s
=> => exporting manifest list sha256:fa907610694596c2b4e1082a967ef427b7ca51ae071b30d4d08291fabd348aab 0.0s
=> => naming to docker.io/library/mi-web-express:latest 0.0s
=> => unpacking to docker.io/library/mi-web-express:latest 0.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ksljmpezopgmauzsalybpwz5y
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker run -d --name test-s
in-puerto mi-web-express
5ce3f2b579d8b559a7c6c22e16b40a213678c07a77834147efd30c6dd3e77cc
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> |
```

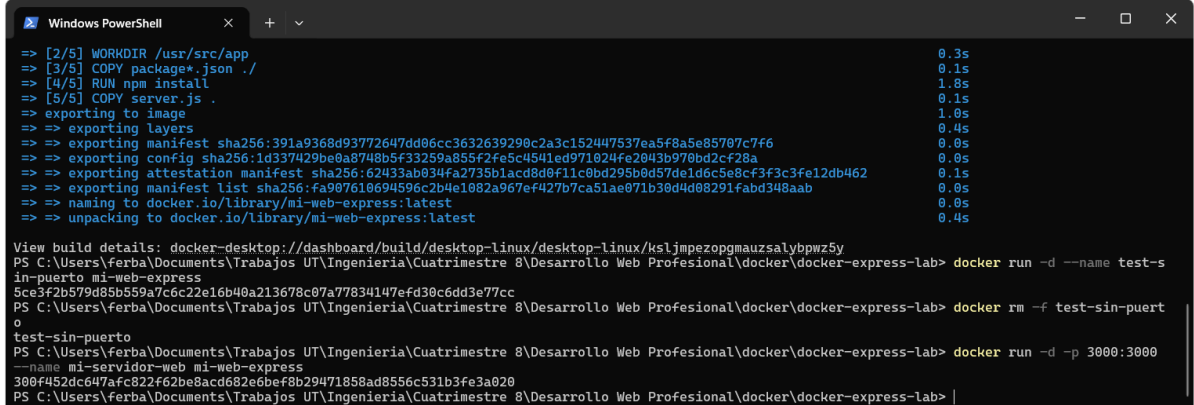
6.

Servidor Express dentro de Docker

Hostname del contenedor: 300f452dc647

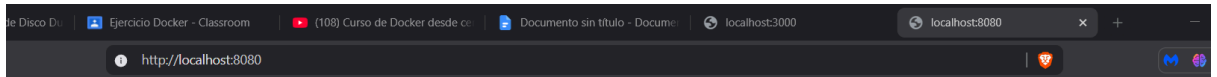
Plataforma OS: Linux (x64)

Versión de Node.js: v22.23.2



```
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker build -t mi-servidor-web .
=> [2/5] WORKDIR /usr/src/app 0.3s
=> [3/5] COPY package*.json ./ 0.1s
=> [4/5] RUN npm install 1.0s
=> [5/5] COPY server.js . 0.1s
=> exporting to image 1.0s
=> => exporting layers 0.4s
=> => exporting manifest sha256:391a9368d93772647dd06cc3632639290c2a3c152447537ea5f8a5e85707c7f6 0.0s
=> => exporting config sha256:1d337429be0a8748b5f33259a855f2fe5c4541ed971024fe2043b970bd2cf28a 0.0s
=> => exporting attestation manifest sha256:62433ab034fa2735blacd8d0f11c0bd295b0d57del6dc5e8cf3f3c3fe12db462 0.1s
=> => exporting manifest list sha256:fa907610694596c2b4e1082a967ef427b7ca51ae071b30d4d08291fabd348aab 0.0s
=> => naming to docker.io/library/mi-web-express:latest 0.0s
=> => unpacking to docker.io/library/mi-web-express:latest 0.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ksljmpezopgmauzsalybpwz5y
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker run -d --name test-s
in-puerto mi-web-express
5ce3f2b579d8b559a7c6c22e16b40a213678c07a77834147efd30c6dd3e77cc
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker rm -f test-sin-puerto
test-sin-puerto
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker run -d -p 3000:3000
--name mi-servidor-web mi-web-express
300f452dc647afc822f62be8acd682e6bef8b29471858ad8556c531b3fe3a020
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> |
```



Servidor Express dentro de Docker

Hostname del contenedor: 690fb7dc695f

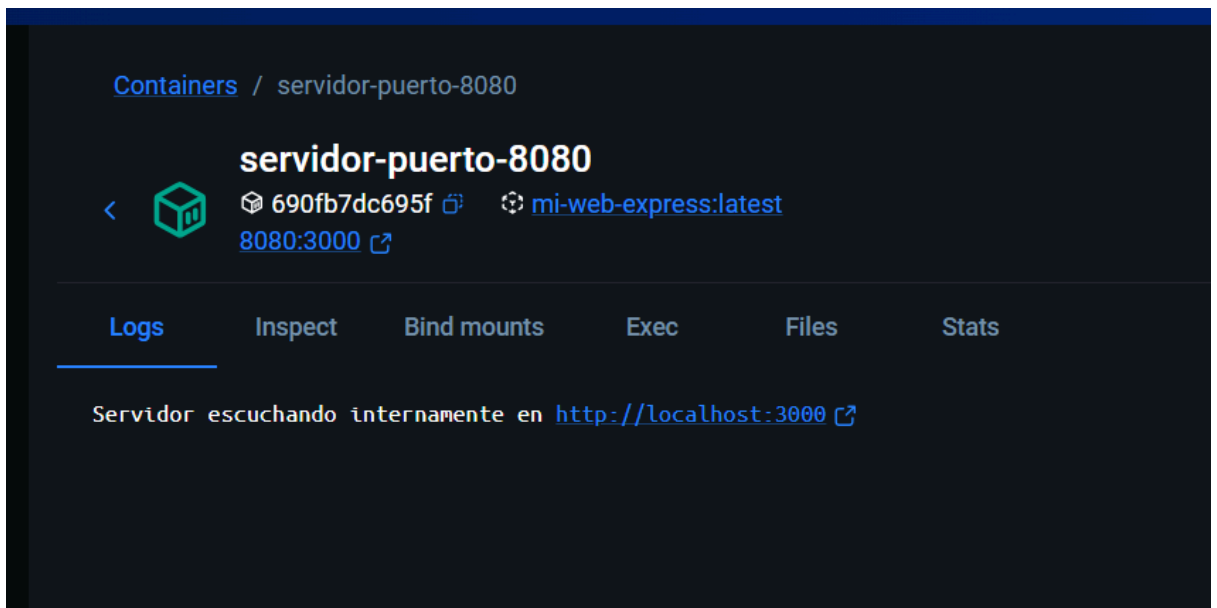
Plataforma OS: Linux (x64)

Versión de Node.js: v22.23.2

```
Windows PowerShell
=> [5/5] COPY server.js .                                0.1s
=> exporting to image                                    1.0s
=> exporting layers                                      0.4s
=> exporting manifest sha256:391a9368d93772647dd06cc3632639290c2a3c152447537ea5f8a5e85707c7f6 0.0s
=> exporting config sha256:1d337429be0a8748b5f33259a855f2fe5c4541ed971024fe2043b970bd2cf28a 0.0s
=> exporting attestation manifest sha256:62433ab034fa2735b1acd8d0f11c0bd295b0d57de1d6c5e8cf3f3c3fe12db462 0.1s
=> exporting manifest list sha256:fa907610694596c2b4e1082a967ef427b7ca51ae071b30d4d08291fabd348aab 0.0s
=> naming to docker.io/library/mi-web-express:latest 0.0s
=> unpacking to docker.io/library/mi-web-express:latest 0.4s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ksljmpezopgmauzsalybpmz5y
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker run -d --name test-sin-puerto mi-web-express
5ce3f2b579d85b559a7c6c22e16b40a213678c07a77834147efd30c6dd3e77cc
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker rm -f test-sin-puerto
test-sin-puerto
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker run -d -p 3000:3000 --name mi-servidor-web mi-web-express
300f452dc647afcf822f62be8acd682e6bef8b29471858ad8556c531b3fe3a020
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker run -d -p 8080:3000 --name servidor-puerto-8080 mi-web-express
690fb7dc695fe7fa1187ad8acd7f88f5936e6315229553a4dfaaef1d2055cec3
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> |
```

7.



Parte 5: Inspección y Monitoreo

```
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
690fb7dc695f   mi-web-express "docker-entrypoint.s..." About a minute Up About a minute 0.0.0.0:8080->3000/tcp, [::]:8080->3000/tcp
servidor-puerto-8080
300f452dc647   mi-web-express "docker-entrypoint.s..." About a minute Up About a minute 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
mi-servidor-web
4f5e86925710   phpmyadmin:latest "/docker-entrypoint..." 3 months ago   Up 15 minutes 0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
pma_gui
9005bdadc6f4   mysql:8.0     "docker-entrypoint.s..." 3 months ago   Up 15 minutes 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp
mysql_db
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> |
```

8.

Ver logs del servidor en vivo:

```
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker logs -f mi-servidor-web
Servidor escuchando internamente en http://localhost:3000
```

9.

Detener y limpiar contenedores:

```
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker stop mi-servidor-  
web servidor-puerto-8080  
mi-servidor-web  
servidor-puerto-8080  
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> docker rm mi-servidor-we  
b servidor-puerto-8080  
mi-servidor-web  
servidor-puerto-8080  
PS C:\Users\ferba\Documents\Trabajos UT\Ingenieria\Cuatrimestre 8\Desarrollo Web Profesional\docker\docker-express-lab> |
```